

Numerical Linear Algebra II: 03 Finite Differences

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The Poisson and Laplace Equations

Poisson's Equation is a widely used partial differential equation (PDE) in physics and engineering (electrostatics, heat conduction, fluid dynamics).

In 1D:

$$-u''(x) = f(x) \quad \text{on } \Omega = (0, 1)$$

In 2D:

$$-\Delta u(x, y) = -\left(\frac{\partial^2 u}{\partial x^2} + \frac{\partial^2 u}{\partial y^2}\right) = f(x, y) \quad \text{on } \Omega = (0, 1) \times (0, 1)$$

If $f = 0$, this is called **Laplace's Equation**. We must also specify **Boundary Conditions** (e.g., $u = 0$ on the boundary $\partial\Omega$, known as Dirichlet boundary conditions).

The Laplace Operator in Mathematical Modeling

The Laplace operator Δu represents *"the difference between the value of a function at a point and its average value in the surrounding neighborhood."*

It is the mathematical engine behind models of **equilibrium** and **diffusion**:

- **Heat Conduction:** The Heat Equation ($\frac{\partial u}{\partial t} = \alpha \Delta u$) models temperature spreading. At steady-state thermal equilibrium ($\frac{\partial u}{\partial t} = 0$), this simplifies to Laplace's equation $\Delta u = 0$.
- **Electrostatics:** Poisson's equation $-\Delta \phi = \frac{\rho}{\epsilon_0}$ describes the electric potential ϕ generated by a charge density ρ .
- **Fluid Dynamics:** The velocity potential of an incompressible, irrotational fluid flow satisfies $\Delta \phi = 0$.
- **Solid Mechanics:** It models the static displacement of elastic strings (1D) and membranes (2D) under physical loads.

Derivation: Taylor Series Expansion

To discretize the continuous domain, we approximate the derivatives using finite differences. We start with the Taylor series expansion of $u(x)$ around a point x_i :

Forward step by $\Delta x = h$:

$$u(x_i + h) = u(x_i) + hu'(x_i) + \frac{h^2}{2}u''(x_i) + \frac{h^3}{6}u'''(x_i) + \frac{h^4}{24}u^{(4)}(x_i) + \mathcal{O}(h^5)$$

Backward step by $\Delta x = -h$:

$$u(x_i - h) = u(x_i) - hu'(x_i) + \frac{h^2}{2}u''(x_i) - \frac{h^3}{6}u'''(x_i) + \frac{h^4}{24}u^{(4)}(x_i) + \mathcal{O}(h^5)$$

Finite Difference for the 1st Derivative

Using the Taylor expansions, we can construct approximations for $u'(x_i)$:

Forward Difference: (using the forward step)

$$u'(x_i) = \frac{u(x_i + h) - u(x_i)}{h} + \mathcal{O}(h)$$

Backward Difference: (using the backward step)

$$u'(x_i) = \frac{u(x_i) - u(x_i - h)}{h} + \mathcal{O}(h)$$

Central Difference: (subtracting the two expansions)

$$\begin{aligned} u(x_i + h) - u(x_i - h) &= 2hu'(x_i) + \mathcal{O}(h^3) \\ u'(x_i) &= \frac{u(x_i + h) - u(x_i - h)}{2h} + \mathcal{O}(h^2) \end{aligned}$$

Finite Difference for the 2nd Derivative

Adding the two Taylor expansions eliminates the odd derivatives:

$$u(x_i + h) + u(x_i - h) = 2u(x_i) + h^2 u''(x_i) + \mathcal{O}(h^4)$$

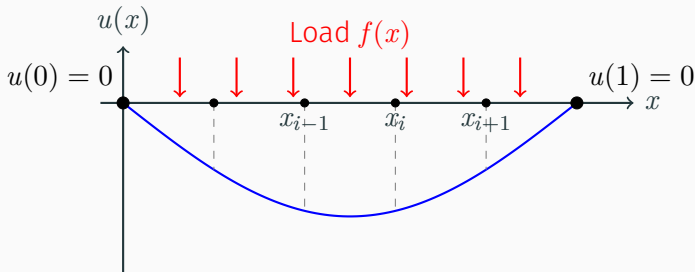
Rearranging for $u''(x_i)$, we get the **Central Difference Approximation**:

$$u''(x_i) = \frac{u(x_i - h) - 2u(x_i) + u(x_i + h)}{h^2} + \mathcal{O}(h^2)$$

This approximation has an error of order $\mathcal{O}(h^2)$, meaning halving the grid spacing h divides the approximation error by 4.

1D Problem: Displacement of a String

The 1D Poisson equation $-u''(x) = f(x)$ can be interpreted as the vertical displacement $u(x)$ of an elastic string stretched between two points, subject to a load $f(x)$.



1D Discretization

Consider a 1D domain $[0, 1]$ discretized into $N + 1$ intervals with spacing $h = \frac{1}{N+1}$.

The grid points are $x_i = ih$ for $i = 0, 1, \dots, N + 1$. Let $u_i \approx u(x_i)$ and $f_i = f(x_i)$.

Applying our approximation to $-u''(x) = f(x)$ at the interior points ($i = 1, \dots, N$):

$$-\frac{u_{i-1} - 2u_i + u_{i+1}}{h^2} = f_i$$

Assuming Dirichlet boundary conditions $u_0 = 0$ and $u_{N+1} = 0$:

$$i = 1 : \quad \frac{2u_1 - u_2}{h^2} = f_1 \quad (\text{since } u_0 = 0)$$

$$i = 2 : \quad \frac{-u_1 + 2u_2 - u_3}{h^2} = f_2$$

$$\vdots$$

$$i = N : \quad \frac{-u_{N-1} + 2u_N}{h^2} = f_N \quad (\text{since } u_{N+1} = 0)$$

1D Matrix Formulation

This system of linear equations can be written in matrix form $\mathbf{A}\mathbf{u} = \mathbf{b}$:

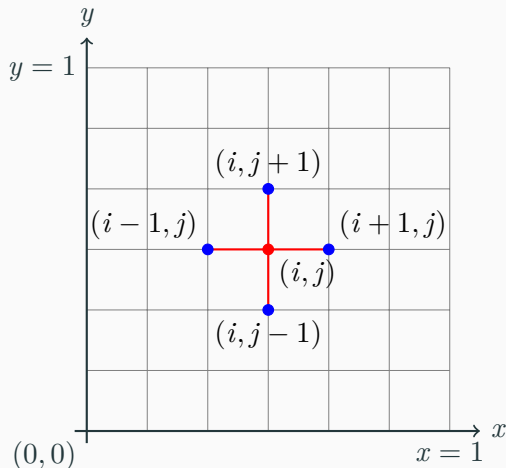
$$\frac{1}{h^2} \begin{pmatrix} 2 & -1 & 0 & \dots & \dots & \dots & 0 \\ -1 & 2 & -1 & 0 & \dots & \dots & 0 \\ 0 & -1 & 2 & -1 & 0 & \dots & 0 \\ \vdots & \ddots & \ddots & \ddots & \ddots & \ddots & \vdots \\ 0 & \dots & \dots & 0 & -1 & 2 & -1 \\ 0 & \dots & \dots & \dots & 0 & -1 & 2 \end{pmatrix} \begin{pmatrix} u_1 \\ u_2 \\ u_3 \\ \vdots \\ u_N \end{pmatrix} = \begin{pmatrix} f_1 \\ f_2 \\ f_3 \\ \vdots \\ f_N \end{pmatrix}$$

The matrix $\mathbf{A}$ is:

- Symmetric Positive Definite (SPD)
- Tridiagonal
- Sparse

2D Problem: Discretization on a Mesh

For the 2D Poisson equation, the domain is a surface (like an elastic membrane). We discretize it using a 2D grid (mesh).



2D Discretization

In 2D, we discretize $\Omega = [0, 1] \times [0, 1]$ using a grid with spacing $h_x = h_y = h$. Let $u_{i,j} \approx u(x_i, y_j)$.

The 2D Laplacian is $\Delta u = \frac{\partial^2 u}{\partial x^2} + \frac{\partial^2 u}{\partial y^2}$. We apply the 1D central difference approximation in both directions:

$$\begin{aligned}\frac{\partial^2 u}{\partial x^2} &\approx \frac{u_{i-1,j} - 2u_{i,j} + u_{i+1,j}}{h^2} \\ \frac{\partial^2 u}{\partial y^2} &\approx \frac{u_{i,j-1} - 2u_{i,j} + u_{i,j+1}}{h^2}\end{aligned}$$

Substituting into $-\Delta u = f$ gives the **5-point stencil**:

$$\frac{4u_{i,j} - u_{i-1,j} - u_{i+1,j} - u_{i,j-1} - u_{i,j+1}}{h^2} = f_{i,j}$$

2D Matrix Formulation

To write $\mathbf{A}\mathbf{u} = \mathbf{b}$, we flatten the $N \times N$ grid into a 1D vector. Grid coordinates (i, j) map to the 1D index k , e.g.,

$$k(i, j) = i + (j - 1)N \text{ (row-by-row)} \quad k(i, j) = j + (i - 1)N \text{ (col-by-col)}$$

This creates a **five-diagonal** matrix with bands $(-1, -1, 4, -1, -1)$. The outer bands are exactly N indices away from the main diagonal:

$$\mathbf{A} = \frac{1}{h^2} \begin{pmatrix} \ddots & & & & & & & & & \\ \dots & -1 & \dots & -1 & 4 & -1 & \dots & -1 & \dots & \dots \\ \dots & \dots & -1 & \dots & -1 & 4 & -1 & \dots & -1 & \dots \\ \dots & \dots & \dots & -1 & \dots & -1 & 4 & -1 & \dots & -1 \\ & \ddots & \ddots & & \ddots & \ddots & \ddots & & \ddots & \ddots \end{pmatrix}$$